

SafeExec

Final Integrated Submission and Demo

A hardened, threat-modeled Python execution sandbox for LLM-agent tool use.

4/4

Fresh W14 checks

smoke, tests, validation, audit

50/50

Docker target-host probes

W8 hardened Docker

49/50

gVisor target-host probes

one timing miss

Hard Stop 7

Student: Zixuan Liang

Advisor: Prof. Khalid Lateef

Instructor: Dr. Majid Shaalan

Prepared: 2026-06-27

Final message

The report, code package, evidence bundle, and demo now tell one evidence-backed story.

Problem: generated code needs a measured boundary.

Risk

LLM agents can turn prompt injection, model error, or excessive autonomy into real process execution.

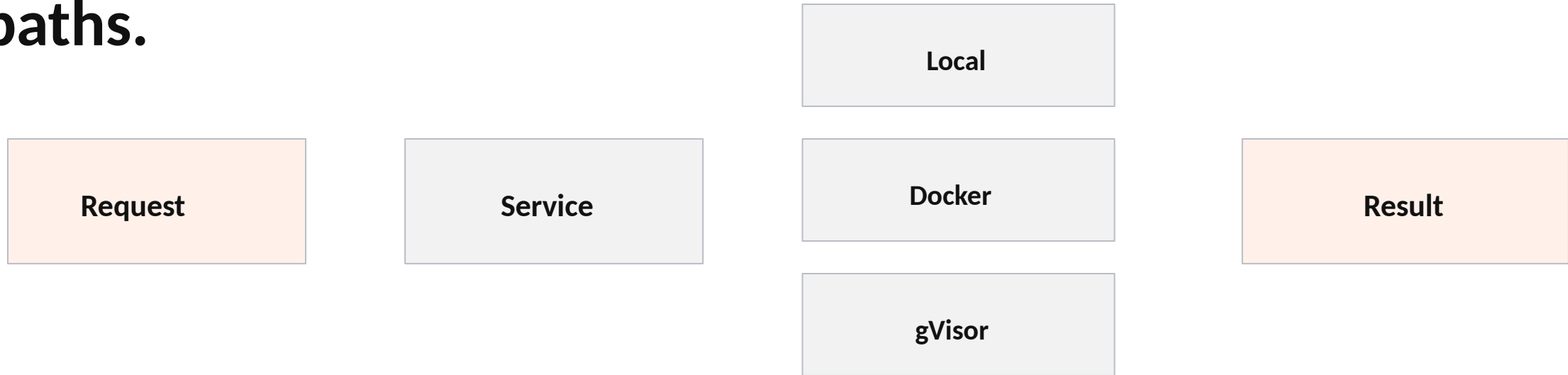
Response

SafeExec keeps scope narrow: Python-only execution, explicit limits, isolated backend paths, and repeatable validation.

Claim

The project does not assert sandbox safety by prose. It reports what the measured boundary currently supports.

Architecture: one service model, three execution paths.



Local is a development smoke path. Docker and gVisor are the containment-relevant paths.

Final artifact surface is compact and traceable.

models.py	request, limits, and result schema
service.py	backend selection and execution entry
backends/docker.py	Docker and gVisor command construction
api/server.py	JSON /health and /execute API shell
run_* scripts	validation, midpoint, final evidence
docs/reproducibility	runbook, environment, data, manifest

The final package includes code, tests, reproducibility docs, evidence logs, and a manifest so the reviewer can inspect more than the narrative.

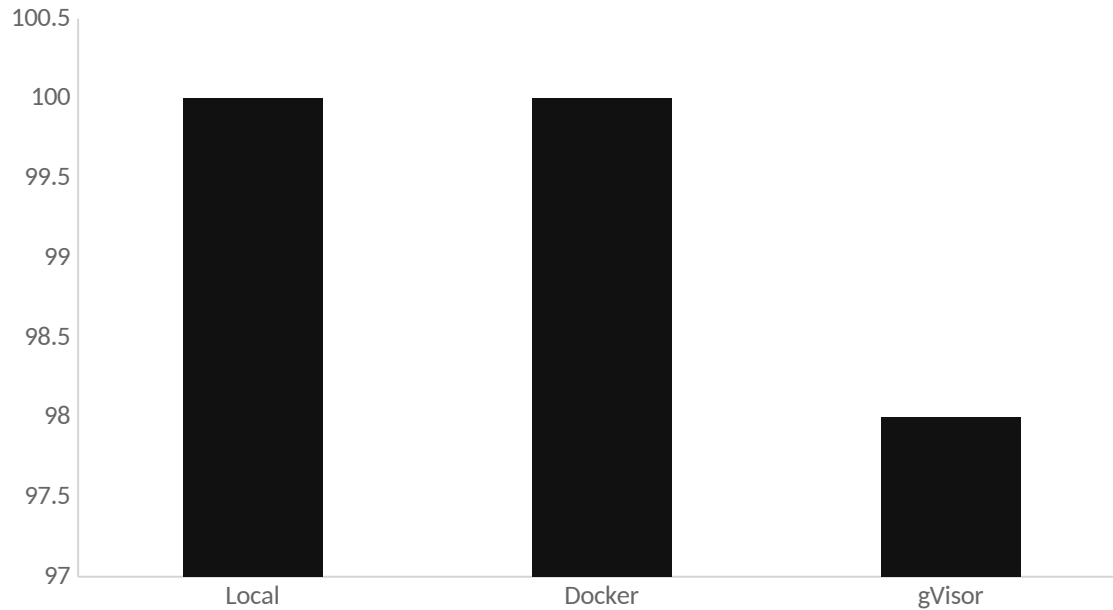
Fresh W14 local evidence passed.

PASS Smoke structured execution result	PASS Unit tests functional test suite	12/12 Validation local cases plus API trace	PASS Repro audit no blocking findings
--	---	---	---

Generated into docs/14-hard-stop-7/evidence/ and archived as final-evidence.zip.

This supports the service/API/reproducibility claims. Container containment evidence remains tied to the Linux target host.

Target-host evidence: Docker stable, gVisor mostly passing.



W8 benchmark summary

Local: 30/30

Docker: 50/50

gVisor: 49/50

Interpretation: gVisor is integrated, but timing needs cold-start and warm-run separation before stronger performance claims.

Final limitations are explicit.

What not to overclaim

- Local backend is not a sandbox.
- gVisor has one current timing miss.
- Corpus depth can keep growing after course closeout.

How the final package handles it

- Preserve Docker/gVisor target-host boundary.
- Report timing caveat honestly.
- Package raw logs and manifests for verification.

Conclusion: SafeExec is a credible graduate-level artifact because its claims are bounded by evidence.

Canvas package: report, artifact, evidence, demo.

- Final-Technical-Report.pdf / .docx
- Final-Integrated-Submission.pdf / .docx
- safeexec-final-artifact-package.zip
- final-evidence.zip and evidence appendix
- Final-Presentation-and-Demo-Deck.pptx / .pdf
- final-presentation-demo-script.md and AI-use log

Record the final walkthrough from the script, then upload the package.